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Applying human factors analysis and classification system for commercial vehicles crashes investigation and critical failure routes analysis

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ABSTRACT

This study aims to develop a reliable and valid Human Factors Analysis and Classification System (HFACS) and Bayesian Network (BN) methodology to understand the causal factors of commercial vehicles (CMV) involved traffic crashes. The HFACS-CMV method has been established using learnings from 100 multi casualty crashes involving at least 10 fatalities. The extreme nature of such crashes ensures the existence of in-depth investigation reports which are necessary to generate sufficient data for HFACS analysis. By analyzing 100 road traffic investigation reports across 28 provinces in China from 2001 to 2021, the research employs odds ratio and BN which is able to quantitatively examine the relationships among contributing factors. The study identifies the highest frequencies of failures in establishing/implementing safety production systems, wrong responses to emergencies, and over speeding across four levels, 12 categories, and 53 sub-subcategories' HAFCS-CMV framework. Numerous associations between the upper and adjacent lower levels are revealed, especially between poor company supervision and government oversight across various subcategories. The HFACS-CMV with BN model highlights a critical failure route: inadequate government oversight leading to poor company supervision or poorly planned operations, resulting in substandard operator conditions and unsafe acts. Identifying associations and failure routes is crucial for developing effective countermeasures. Specific outcomes are limited by the crash reports used which were not generated with HFACS in mind, future research using HFACS informed crash reporting systems would be beneficial. However, the successful application of the method demonstrates the efficiency and applicability of HFACS-CMV as a robust method for understanding causal factors of road crashes.

1. Introduction

The Global Status Report on Road Safety 2023 reveals that 1.19 million people die each year in road traffic crashes (WHO, 2023). In 2019, China experienced 62,763 fatalities due to road crashes. According to the 2019 China Traffic Crash Yearbook, commercial vehicles (CMV) account for approximately 30 %-40 % of fatal crashes, with freight trucks and coaches being the primary contributors (Ministry of Transport of the People's Republic of China, 2019) (Ministry of Public Security of the People's Republic of China, 2018). Particularly, crashes resulting in ten or more fatalities have significant consequences, affecting life safety, causing economic losses, and leading to social and environmental issues. The CMVs refer to vehicles engaged in road transport operations, including road passenger transport vehicles and road freight transport vehicles. (Ministry of Transport of the People's Republic of China, 2022). The severity and wide-ranging impacts of CMV crashes underscore the urgent need for effective road safety measures to reduce fatalities and mitigate their effects.

Numerous studies have used safety analysis to identify factors affecting crashes, showing that 57 % are directly caused by human factors and 90 % involve human factors (Y. Zhang et al., 2016). However, most of experts involved in crash research and analysis would

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argue that merely attributing crashes to human error is overly simplistic (Filtness et al., 2014). Instead, in-depth crash analysis methods are required to gain a deeper understanding of the underlying issues and truly make a difference in safety. Especially, the causal factor of CMV involved traffic crash is characterized by complexity and multi-level features. Aspects such as safety culture or climate should be introduced, as they contribute to increased transport safety (Nævestad et al., 2021). Human factors should not be simplistically equated with human error or unsafe driving behaviors, rather investigation efforts should be extended to find out what exactly those human factors are.

Despite significant technological advancements in road traffic crash investigations globally, a common deficiency persists: inadequate exploration of corporate management, government regulation, and organizational factors. This gap limits the ability to fully understand the root causes of crashes and hinders the development of more effective preventive measures. To elevate traffic safety, it is crucial to conduct deeper and more systematic studies on these elements. Systematic models of road traffic crashes, quantitative measurements of interactions among causal factors, and comprehensive research methods are needed. The Human Factors Analysis and Classification System (HFACS) developed within the aviation industry, offers a valuable framework for this purpose (Wiegmann & Shappell, 2017). HFACS assesses operator preconditions, environmental factors, unsafe supervision, and organizational influences.

Furthermore, although HFACS is a useful tool to explore the possible failure modes in risk analysis, it can only identify the potential failure routes, because of the lack of quantitative risk assessment techniques. The integration of HFACS with quantitative data enables a thorough assessment of the influence and hazards posed by these factors on occurrences, consequently enhancing our comprehension of causal mechanisms of incidents and their prevention strategies. Existing causation analysis is limited by an over-reliance on statistical analysis, which often overlooks complex interactions and deeper causal relationships, limiting a comprehensive understanding of crash causes and effective safety interventions. While methods like grey correlation analysis have been applied in some studies, techniques such as Bayesian Networks (BN) are underutilized in CMV traffic crashes.

For these reasons, this study first aims to adapt and design HFACS specifically for traffic crashes involving commercial vehicles. Validated HFACS-CMV will then be applied to multiple crash investigation reports to extract key causal factors. Finally, quantitative methods such as Bayesian Networks and HFACS will be utilized to identify critical failure modes and propose targeted preventive measures. This approach provides a new method for human factors analysis in CMV traffic crashes, addressing shortcomings in traditional crash investigation and analysis. Simultaneously, the study will reveal interactions among factors at various levels, identify key factors contributing to crashes, and propose appropriate preventive measures.

2. Literature review

2.1. In-depth investigation of road traffic crashes

In-depth investigation of road traffic crashes is essential for enhancing traffic safety, reducing crash rates, and improving survival rates for road users. In many countries, the activities are conducted by police collision investigators. However, police investigations primarily focus on establishing blame for the judiciary, often overlooking causation factors that could prevent future crashes (Gooding & Box, 2022). Moreover, investigation of road traffic crashes remains predominantly on immediate technical causes and driver behavior (Rolison et al., 2018). The National Transportation Safety Board (NTSB) conducts in-depth investigations into civil aviation crashes, thoroughly examining not only the technical causes but also underlying factors related to organizational management, training programs, and regulatory oversight. This disparity highlights significant shortcomings in how road traffic crashes are investigated, particularly in their failure to scrutinize the safety management systems of transportation companies, the thoroughness of training protocols, and the enforcement of regulations.

Although the proximate causes of the crashes were the unsafe driving behavior, responsibility for the crashes cannot simply be attributed to the driver (Newnam & Goode, 2015). Research has increasingly recognized that organizational and regulatory shortcomings play a critical role in crash occurrence. For instance, Zhang et al. (2023) identified inadequate safety management as the root cause of most bus and coach crashes. Similarly, studies on road freight safety underscore how systemic failures, like poor safety policies, contribute significantly to crashes, emphasizing the need for crash investigations to address these organizational and systemic factors (Newnam & Goode, 2015). Therefore, it is necessary to identify human factors behind CMV crashes aids in enhancing the effectiveness of traffic crash prevention measures, particularly in addressing systemic management problems and establishing long-term regulatory frameworks.

However, despite the recognition of systemic factors, there remains a lack of a standardized methodological framework to systematically assess the impact of human factors in CMV crashes. While crashes often stem from driver errors, the classification of contributing factors is still inconsistent, making it difficult to establish effective preventive measures (Stanton & Salmon, 2009). Supporting this perspective, Newnam et al. (2017) analyzed Australian Coronial inquests into road freight crashes and found that while these investigations identified multiple contributing factors (e.g., driver behavior, organizational policies, and regulatory gaps), their recommendations often lacked a systems-thinking approach. This highlights the critical need for systemic reform in road CMV transportation, including the development of frameworks that capture the complex interactions within the system to inform more holistic prevention strategies.

In China, according to the "Regulations on the Reporting, Investigation, and Handling of Work Safety Crashes", crashes resulting in ten or more fatalities are investigated by the provincial government of the area where the crash occurred, while those with thirty or more fatalities are investigated by a special team organized by the State Council or authorized departments (State Council, 2008). For each crash, a road traffic crash investigation team is established, comprising responsible officials from the Public Security Department, the Department of Transport, the Department of Emergency Management, the Ministry of Industry and Information Technology of the People's Republic of China, the Market Regulation Administration, and the relevant local government. Additionally, relevant experts are invited to participate in the investigation. Through on-site inspections, sample testing, evidence collection, document review, personnel interviews, and expert consultations, the investigation aims to ascertain the sequence of events, causes, casualties, and direct economic losses associated with the crash. The nature and liability of the crash are determined, and any issues related to relevant entities and individuals are identified. These investigations result in detailed reports, each over 10 pages (more than 3,500 words), which provide in-depth information about the crash process, rescue operations, and relevant details such as the involved units, responsible drivers, vehicles, road conditions, maintenance issues, casualties, and economic losses. This comprehensive reporting offers a valuable opportunity for conducting system-based investigations and understanding the root causes of crashes.

2.2. Human factors analysis and classification system

Crashes occur due to complex mechanisms, and various models have enhanced our understanding of them. Classic theoretical models in behavioral science include the Hawkins SHEL model, "Swiss Cheese" model, and the Reason's model. As Reason (1997) noted, while active failures by operators directly affect safety, many crashes stem from latent failures originating in higher levels of management and organizational levels. Among these, the HFACS is a widely used framework for

crash analysis basing on Reason and the 'Swiss Cheese' model, which situate unsafe acts within an organizational context (Wiegmann & Shappell, 2017). In the HFACS system, human factors issues are categorized into four levels: Organizational Influences, Unsafe Supervision, Preconditions for Unsafe Acts, and Unsafe Acts as Fig. 1. These levels are interconnected in a causal chain, where organizational influences shape supervisory practices, which in turn create the conditions for unsafe acts. This structure helps to identify how various factors at different levels contribute to crashes, from organizational policies to individual behaviors. HFACS as a comprehensive approach helps investigators and analysts identify system deficiencies more clearly.

Research on HFACS applications can be divided into three primary types. The first type used the original HFACS framework to analyze crash causation, with its earliest and most extensive applications in aviation crash analysis (Zhou et al., 2018). For instance, studies on coal mine crashes have revealed a high incidence of unsafe behaviors and insufficient supervision, showcasing the model's adaptability beyond the aviation sector (Wachter & Yorio, 2014). The second type integrates other crash causation methods with the HFACS framework to broaden the analysis. For example, combining HFACS with Fault Tree Analysis enables the identification of fundamental collision risk factors by statistically analyzing crash reports and expert judgments (M. Zhang et al., 2019). Although HFACS and its derived frameworks have been extensively studied in aviation and maritime contexts, their application to road traffic crash analysis remains limited.

The third type focuses on revising and optimizing the HFACS framework to enhance its applicability and specificity across various fields. Although initially designed for aviation, HFACS has since been adapted to sectors such as maritime, rail, and mining (Li et al., 2019). However, the framework's reliability remains a concern, as the coding scheme's consistency significantly influences the validity of findings (Ergai et al., 2016). In a review of 111 studies, Cohen et al. (2015) reported that only 14 studies provided HFACS reliability data, underscoring the need for rigorous validation. Li and Harris tested 18 causal factors in civil aviation crashes, finding Cohen's Kappa values indicating moderate reliability (Li & Harris, 2005). Additionally, Olsen and colleagues find many of the tools available have demonstrated poor reliability of coding, making their validity and usefulness questionable (Olsen & Williamson, 2017). To improve its utility for analyzing commercial vehicle-related traffic crashes, HFACS requires adjustments specific to these crashes and a thorough reliability assessment.

2.3. HFACS with quantitative methods for commonality analysis

Integrating HFACS with quantitative methods can deepen its analysis, decipher common crashes causes and broaden its application. For example, Akhtar and Utne's (2014) study demonstrated the significant impact of fatigue on maritime grounding probability by combining HFACS with correlation analysis for quantitative evaluation (Akhtar & Utne, 2014). Feng et al. (2018) integrated HFACS with grey correlation analysis to identify key factors in coal mining casualties. Their findings showed that "unsafe acts" had the highest grey relational grade with crash casualties, while "inadequate supervision" was the category most closely linked to these crashes' casualties. Harris and Li used neural networks with HFACS to predict unsafe acts (level 1) based on preconditions for unsafe acts (level 2) in military aircraft crashes, achieving an average classification accuracy of 74 % (Harris & Li, 2019).

BN is capable of revealing the interrelations and dependencies among model parameters, providing a clearer understanding of complex problem (Trucco et al., 2008). This approach has been successfully applied in fields such as medical diagnosis, statistical decision-making, expert systems, and learning prediction, demonstrating its power for reasoning under uncertainty (Cheng et al., 2019). The structure of BN is divided into qualitative and quantitative components. At the qualitative level, a Directed Acyclic Graph is employed to represent the dependencies and independencies between two sets of variables. At the quantitative level, Conditional Probability Tables are used to describe the degree of dependency among variables. Consequently, BN is particularly well-suited for integrating with the HFACS framework, which proposes a causal chain from organizational policies to individual behaviors, to conduct quantitative analysis of crashes causation (Wang et al., 2013). Furthermore, compared to other quantitative crash

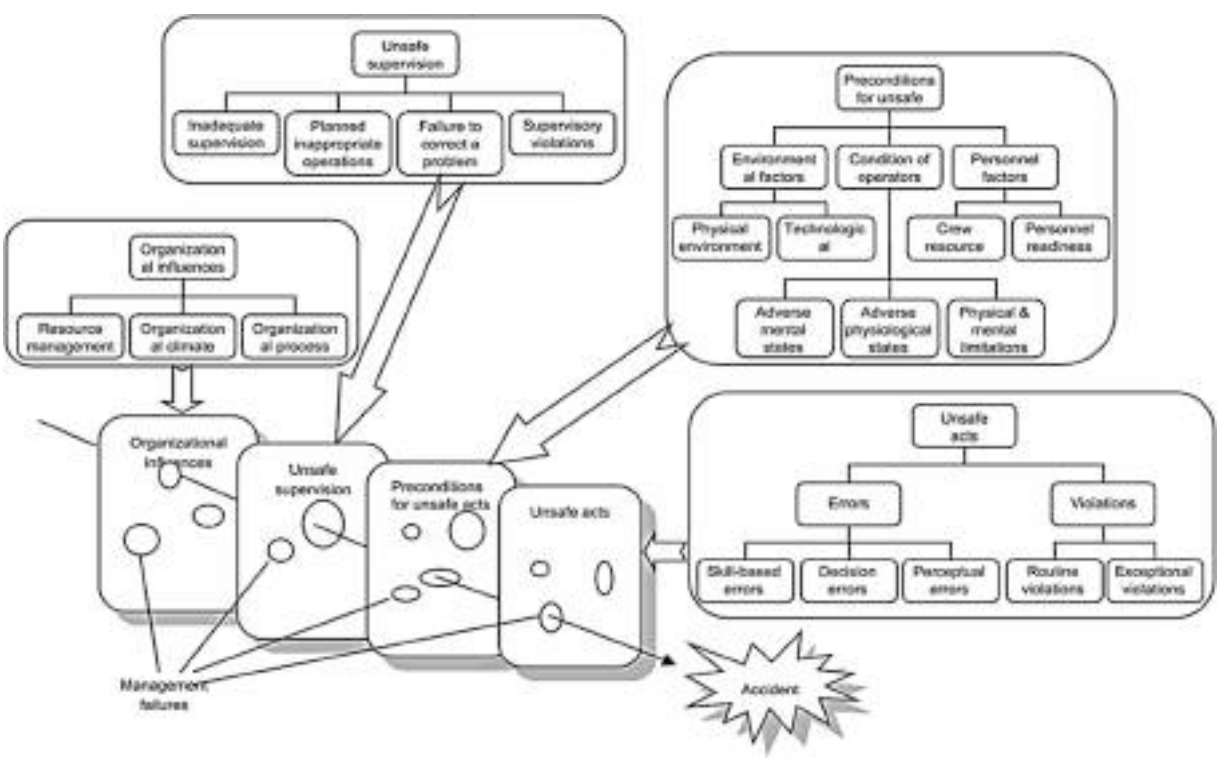

Fig. 1. HFACS framework (Wiegmann & Shappell, 2017).

analysis methods, such as the analytic hierarchy process, event tree analysis, and fuzzy comprehensive evaluation, BN offers the significant advantage of probabilistic reasoning (Romeijn et al., 2016). This allows for more precise and nuanced analysis of crash causation.

The combination of the HFACS framework and the BN model has been explored in previous studies (Dragović & Onjia, 2007), while some limitations are also remained to be solved. For instance, Zhou et al. proposed a hybrid HFACS-BN model to investigate the impact of human factors on aviation crash risk through a questionnaire survey, which introduces a degree of subjectivity (Zhou et al., 2018). Simultaneously, most existing studies have focused on aviation and marine systems, leaving a gap in comprehensive analysis related to road crashes and risks. Detailed countermeasures and recommendations to reduce road crash risks often fall short in depth and sufficiency. To addressing these limitations, the application of HFACS-BN models to road traffic crashes will be expanded and insightful countermeasures based on robust data analysis will be developed.

In conclusion, the HFACS provides a valuable framework for understanding human factors in crashes. However, it has not yet been applied to road transportation. Adapting HFACS to the context of CMV traffic crashes and integrating it with quantitative methods can significantly improve the analysis and prevention of these crashes.

3. HFACS-CMV framework

3.1. HFACS-CMV design

We used the original HFACS framework described by Wiegmann and Shappell (Wiegmann & Shappell, 2017). Considering the characteristics of HFACS and road traffic crashes, the classification framework of CMV crashes was designed with reference to the Model Minimum Uniform Crash Criteria (MMUCC) from the U.S. National Highway Traffic Safety Administration and the Road Traffic Crash Information Collection Table (CICT) in China.

Subsequently, a research team composed of three experts holding doctoral degrees in transportation was established, comprising university scholars and personnel from transportation enterprises. They have all undergone qualitative analysis training in previous research or projects, and possess extensive coding experience. They also underwent at least 10 h of joint training on the HFACS framework to ensure a thorough and accurate understanding of each category and sub-category. The experts discussed their disagreements in the designing process until they reached a consensus. After extensive discussions on the coding scheme, the initial version of HFACS-CMV was developed (Fig. 2).

Firstly, the four levels of the HFACS framework were retained, with modifications made to the categories as following:

1. Level 4 (Organizational Influences)

Replaced "Organizational Process" with "Inadequate Oversight of Government". The original "Organizational Process" is suited to aviation, where management, policies, procedures, and resource allocation are critical. In road transportation, more diverse operational structures (e.g., private transport companies, government agencies, various driver types) require a focus on governmental oversight.

2. Level 3 (Unsafe Supervision)

Removed "Supervisory Violation". In aviation, "Supervisory Violation" refers to instances where supervisors fail to enforce rules or procedures effectively. However, in road transportation, supervision is often less centralized, with various private companies, government agencies, and independent drivers involved.

3. Level 2 (Preconditions for Unsafe Acts)

Removed "Personnel Factors". "Crew Resource" and "Personnel Readiness" are specific to aviation and do not apply to road transportation. Elevated "Physical Environment" and "Technological Environment" to main categories to emphasize the role of vehicle, road, and

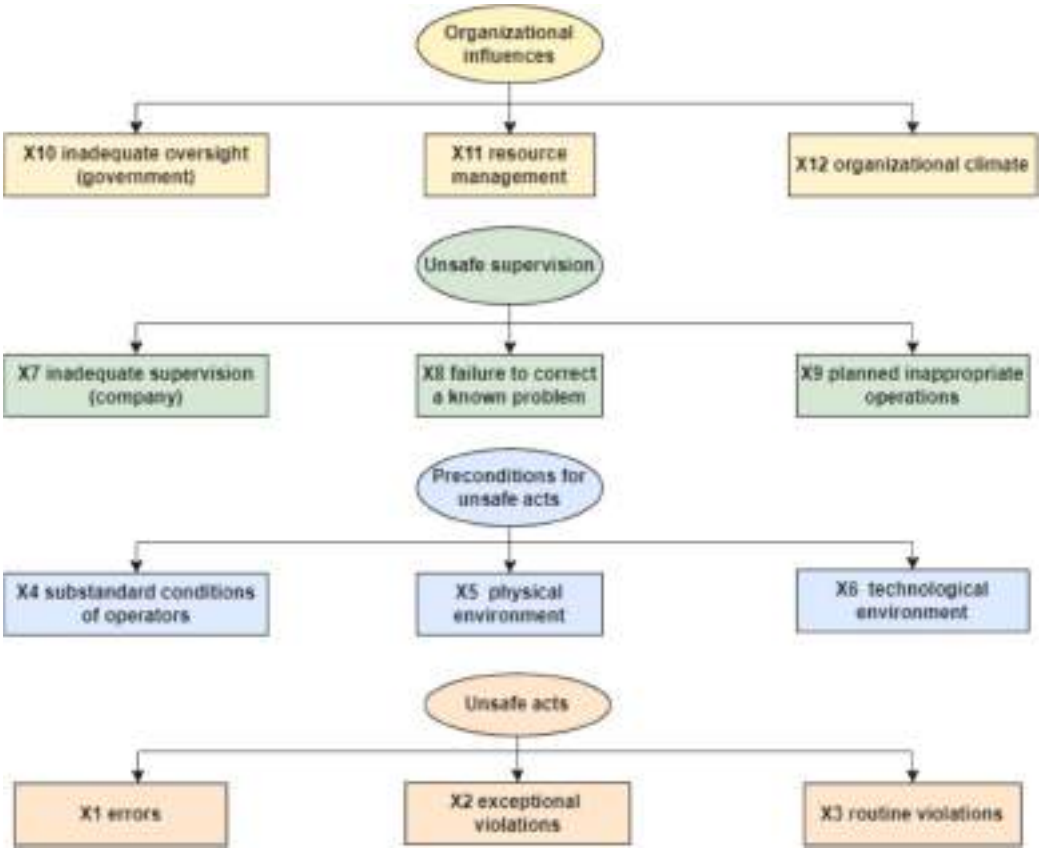

Fig. 2. Levels and categories of HFACS-CMV.

environmental factors in crashes.

4. Level 1 (Unsafe Acts)

Elevated "Routine Violation" and "Exceptional Violation" to main categories. This better reflects CMV-specific violations, such as illegal operations and invalid licenses, that are often characterized by routine violations where drivers knowingly break the law.

Secondly, the revision of sub-categories can be divided into two scenarios: one is consistency and similarity in aviation and road, and the other is differences between aviation and road. In the first scenario, identical and similar sub-categories are retained in HFACS-CMV. For the second scenario, directly remove the sub-categories specific to the aviation field and add sub-categories specific to the road field (shown in Appendix A).

3.2. Reliability and validity testing

To evaluate the inter-rater, intra-rater reliability, and validity of HFACS-CMV. Another five experts who did not participate in the initial version design were invited to this study. They are holding doctoral degrees in transportation, possess over five years of research or practical experience in road safety and human errors and received training in classifying crashes data with the HFACS. Two crash reports were systematically selected from our database of 100 multi-casualty crashes: one inter-vehicle crash (Case A) demonstrating complex organizational and supervisory failures, and one single-vehicle crash (Case B) highlighting individual operator conditions and preconditions. This selection enables comprehensive testing of the HFACS-CMV framework's analytical robustness across all classification levels.

The steps for testing reliability, validity, and optimizing HFACS-CMV were as Fig. 3.

To verify the reliability of the HFACS-CMV using Fleiss Kappa Values (SPSS 27.0), with 0.00 to 0.20 considered slight agreement, 0.21 to 0.40 fair agreement, 0.41 to 0.60 moderate agreement, 0.61 to 0.80 substantial agreement, and 0.81 to 1.00 almost perfect agreement (Plascencia et al., 2021).

The inter-rater and intra-rater reliability measures were determined by calculating Fleiss Kappa Values as **Equation (1)**:

$$K = \frac{\overline{P} - \overline{P_e}}{1 - \overline{P_e}} \tag{1}$$

where $\overline{P}$ is the mean observed proportion of agreement among raters, and $\overline{P_e}$ is the mean expected proportion of agreement by chance.

The mean observed proportion of agreement $\overline{P}$ is calculated using **Equation (2)**:

$$\overline{P} = \frac{1}{N}\sum_{i=1}^{N} P_i \tag{2}$$

where N is the total number of items being rated, and P_i is the proportion of times the raters agreed on the i th item, calculated as **Equation (3)**:

$$P_i = \frac{1}{n(n-1)}\sum_{j=1}^{k} n_{ij}(n_{ij}-1) \tag{3}$$

where n is the number of raters, k is the number of possible categories, and n_{ij} is the number of raters who assigned the i th item to the j th category.

The Kappa value for Case A was 0.78 ($P < 0.001$) and for Case B was 0.83 ($P < 0.001$), indicating substantial inter-rater reliability for HFACS-CMV. The consistency of coding results for each expert, conducted twice, is shown in Table 1. The Kappa values for all experts exceeded 0.6 ($P < 0.01$), confirming the acceptable intra-rater reliability of HFACS-CMV.

Questionnaire was measured on a five-point Likert scale to verify validity with higher scores indicating greater comprehensiveness and accuracy of the level classifications. Table 2 presents the results of the questionnaire used to assess whether the sub-categories and categories in HFACS-CMV had reflected the measured concepts comprehensively and appropriately. The results, shown as mean values ± standard deviations, indicated that scores at each level were consistently above 4, demonstrating good validity for HFACS-CMV. However, scores for higher-level tended to be lower than those for lower-level, suggesting challenges in accurately delineating and articulating higher-level categories.

4. Frequencies of sub-categories

4.1. Data source

The crash data in this study were sourced from official Road Traffic Crash Investigation Reports published by China's Ministry of Emergency Management and provincial governments. Only crashes where commercial motor vehicle (CMV) drivers were assigned full or primary responsibility were included. All selected crashes resulted in 10 or more fatalities, with investigation reports issued by provincial or State Council authorities to ensure comprehensive documentation of causal factors and liability (State Council, 2008). These crashes occurred between 2001 and 2021 across 28 provinces, municipalities, and special

Table 1
Intra-rater reliability.

	Expert 1	Expert 2	Expert 3	Expert 4	Expert 5
Case A	0.72**	0.61**	0.78**	0.76**	0.67**
Case B	0.86**	0.62**	0.88**	0.87***	0.79**

***$P < 0.001$, **$P < 0.01$.

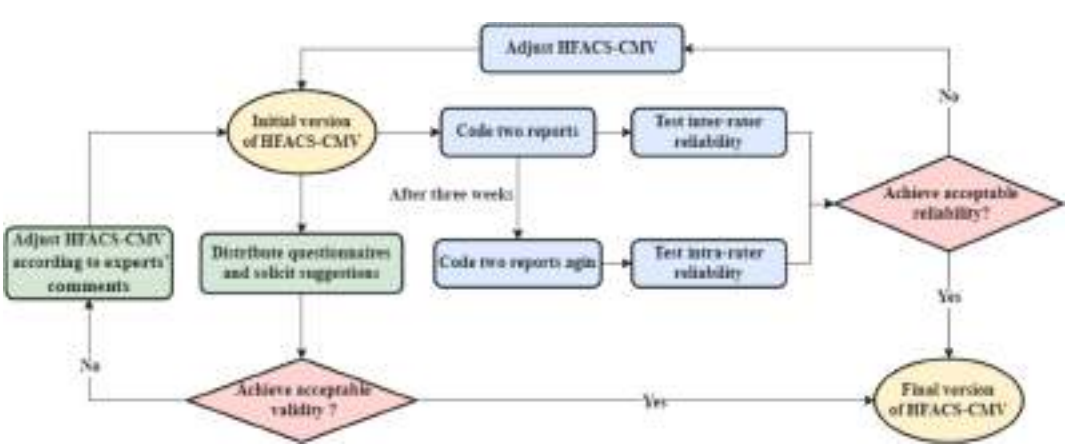

Fig. 3. HFACS-CMV adjustment process.

Table 2
Content validity questionnaire.

	Unsafe acts	Preconditions for unsafe acts	Unsafe supervision	Organizational influences
1. Is the statement clear and concise?1 (Extremely unclear) − 5 (Very clear)	4.50 ± 1.00	5.00 ± 0.00	4.25 ± 0.50	4.50 ± 0.58
2. Does it cover all possible variables and factors?1 (Not covered) − 5 (Completely covered)	4.50 ± 0.57	4.75 ± 0.50	4.25 ± 0.50	4.25 ± 0.95
3. Is it easy to understand and operate? 1 (Not easy to understand) − 5 (Very easy to understand)	4.75 ± 0.50	4.75 ± 0.50	4.50 ± 0.58	4.00 ± 0.82
4. Is the scope too broad?1 (Too broad) − 5 (Appropriate scope)	4.25 ± 0.50	4.75 ± 0.50	4.25 ± 0.50	4.00 ± 0.82
5. Is the scope too narrow?1 (Too narrow) − 5 (Appropriate scope)	4.00 ± 0.82	4.25 ± 0.50	4.00 ± 0.82	4.25 ± 0.50

administrative regions. There are 11 reports from before 2008 and 89 reports from 2009 onwards. The investigative report (over 3,500 words/ 10 pages) systematically documents crash dynamics and post-crash responses through seven analytical dimensions. Each report comprises the following key components to facilitate comprehensive crash reconstruction and causal analysis: vehicle profile, driver background, roadway characteristics, meteorological conditions, corporate entities information, regulatory authority review, issues identified. We have carefully reviewed all reports, which provide comprehensive information.

The analysis reveals that these crashes resulted in 1,583 deaths and 1,930 injuries. These crashes were categorized into inter-vehicle crashes (56 %) and single-vehicle crashes (44 %), which occurred in various locations, including expressways (37 %), primary highway (4 %), secondary highway (26 %), tertiary highway (16 %), quaternary highway (8 %) and non-classified highway (9 %). The primary crashes patterns were vehicle rollover (31 %) and collision (48 %). Among collision crashes, side impacts constitute the highest proportion at 24 %, followed by frontal collisions at 14 %. Among the commercial vehicles involved, large passenger vehicles (seating capacity $\geq$ 20) accounted for the highest proportion (41 %), followed by heavy freight vehicles (gross vehicle weight $\geq$ 12 tons, 33 %), medium passenger vehicles (seating capacity 10–19, 14 %), small passenger vehicles (seating capacity $\leq$ 9, 9 %), and light freight vehicles (gross vehicle weight $<$ 4.5 tons, 3 %).

4.2. Data coding

Three experts, trained in qualitative analysis through previous studies or projects and possessing substantial experience in coding, developed the initial version of HFACS-CMV for this research (detailed in Section 3.1). They have a thorough understanding of each category and subcategory within HFACS-CMV. The experts coded 100 in-depth crash investigation reports using HFACS-CMV, assessing the presence (coded as 1) or absence (coded as 0) of each subcategory in the narrative of the reports. For instance, when the reports identified a driver's blood alcohol content as above the legal limit, all three experts agreed to code the subcategory "drunk driving" as 1. If any discrepancies arose based on the report content, the experts convened to reach a consensus.

The coding process followed a structured approach, starting with the identification of direct and indirect causes outlined in the crash investigation reports. Firstly, conduct preliminary coding of crash causes based on issues identified in reports, then separately code Level 2, Level 3, and Level 4 based on investigative details, as summarized below:

Issues identified in reports

Direct Causation: Technical failures (vehicle/road), operational errors, or severe weather impacts.

Driver-Related Issues: Deficiencies in qualifications, procedural violations, or behavioral errors contributing to the crash.

Corporate-Related Issues: Failures in safety management, inadequate training, or non-compliance with regulatory requirements.

Regulatory Authority-Related Issues: Gaps in enforcement or coordination by relevant authorities, such as public security or transportation agencies.

Accountability Recommendations: Proposed actions for holding responsible individuals or entities accountable, including penalties, operational suspensions, or policy reforms.

Level 2: Investigative details supported coding for preconditions for unsafe acts

Driver Background: Demographic data (age, gender), driving experience, professional credentials, and compliance history (traffic violations, disciplinary records).

Vehicle Profile: Technical specifications (model, manufacturer), maintenance history, and operational documentation (insurance status, traffic violations).

Roadway Characteristics: Infrastructure details (road classification, design, pavement conditions), traffic control systems (signage, road markings), and environmental factors (sight distance, obstructions).

Meteorological Conditions: Quantitative data (temperature, precipitation, wind conditions) and temporal patterns during the pre-crash and crash phases.

Level 3: Investigative details supported coding for unsafe supervision

Corporate Entities Information: Organizational details (timeline, capital structure), safety management systems (maintenance protocols, driver training), and regulatory compliance (business licensing, operational authorization).

Level 4: Investigative details supported coding for organizational influences

Regulatory Authority Review: Policy enforcement, risk control mechanisms (e.g., overload or speeding management programs), and public safety initiatives (traffic law enforcement, educational campaigns).

4.3. Frequencies of sub-categories

The final HFACS-CMV causative factors into four levels, 12 categories, and 53 sub-categories, labeled as X1, X2, X3, etc., with subcategories denoted by suffixes such as a, b, c as shown in Table 3.

All reports detailed unsafe acts and preconditions for unsafe acts, but 11 reports did not indicate unsafe supervision by the company, and 16 reports did not show organizational influences. Notably, four reports lacked both unsafe supervision and organizational influences, suggesting that higher-level factors were sometimes overlooked in crash investigations. While there appears to be variance in the frequencies across different sub-categories, due to we focus on severe crashes (those involving ten or more fatalities) which were relatively rare compared to less severe crashes. The severe crashes often involve complex, multi-

Table 3a
HFACS-CMV and frequencies of level 1 (unsafe acts).

Categories	Sub-categories	Frequencies
X1 errors	x1a poor driving skills	29
	x1b wrong response to emergency	56
	x1c misjudged distance/speed	1
	x1d failure to identify abnormalities	14
	x1e inadequate distance from the preceding vehicle	5
X2 exceptional violations	x2a over speeding	55
	x2b low speeding	1
	x2c overloading/overcrowding	48
	x2d operations in violation of traffic rules	3
	x2e failure to yield	12
	x2f illegal overtaking/lane occupation/ lane changing	17
	x2g driving on the wrong side of the road	6
	x2h running a red light	0
	x2i illegal transportation of people/goods	7
X3 routine violations	x3a damaging surveillance	2
	x3b illegal operation	14
	x3c invalid license	15

Table 3b
HFACS-CMV and frequencies of level 2 (preconditions for unsafe acts).

Categories	Sub-categories	Frequencies
X4 substandard conditions of operators	x4a passengers not wearing seatbelt	8
	x4b drug driving	1
	x4c drunk driving	2
	x4d distracted driving	7
	x4e fatigue driving	12
	x4f unfamiliar with road conditions	5
	x4g medical condition	0
X5 physical environment	x5a weather-related road conditions	24
	x5b adverse weather conditions	53
X6 technological environment	x6a road issues	28
	x6b vehicle issues	37

factorial interactions that are less pronounced in lower-severity crashes, thereby providing critical insights into high-risk scenarios and systemic vulnerabilities.

In Level 1 (Unsafe Acts), the most frequent sub-categories were wrong response to emergency (x1b, 56 occurrences), over speeding (x2a, 55), and overloading/overcrowding (x2c, 48) (Table 3a).

For Level 2 (Preconditions for Unsafe Acts), adverse weather conditions (x5b, 53) and vehicle issues (x6b, 37) were the most frequent (Table 3b).

In Level 3 (Unsafe Supervision), the top sub-categories were failure to establish/implement a safety production system (x7a, 62), insufficient safety training for employees (x7c, 45), and insufficient vehicle inspections (x7d, 34) (Table 3c).

At Level 4 (Organizational Influences), inadequate supervision of companies (x10c, 51), inadequate site monitoring (x10g, 45), and inadequate police/patrol supervision (x10d, 42) were the most sub-categories (Table 3d).

We employed the Mann-Whitney *U* test to compare the differences in HFACS-CMV Sub-categories distribution among road types (highways vs. expressways), crashes patterns (inter-vehicle vs. single-vehicle), and vehicle types (passenger vehicles vs. freight vehicles). The results showed no significant differences ($p > 0.05$), hence, this study did not regroup the 100 crashes based on road types, crashes patterns, or vehicle types for subsequent analysis (see Appendix B).

Table 3c
HFACS-CMV and frequencies of level 3 (unsafe supervisions).

Categories	Sub-categories	Frequencies
X7 poor supervision (company)	x7a failure in establish/implement safety production systems	62
	x7b not established driver profiles	10
	x7c insufficient safety training for employees	45
	x7d insufficient vehicle inspections	34
	x7e inadequate management of affiliated vehicles	26
	x7f inadequate on-route monitoring	33
X8 failure to correct a known problem	x8a failure to rectify and reinspect safety hazards	8
	x8b failure to identify an at-risk driver	8
	x8c failure to correct illegal behavior	16
	x8d failure to identify an at-risk vehicle	12
X9 planned inappropriate operations	x9a inadequate pre-departure safety inspection	11
	x9b improper staffing	4
	x9c transportation tasks not in accordance with rules & regulations	7

Table 3d
HFACS-CMV and frequencies of level 4 (organizational influences).

Categories	Sub-categories	Frequencies
X10 inadequate oversight (government)	x10a ineffective implementation of rules and regulations	29
	x10b inadequate road maintenance	10
	x10c inadequate supervision of companies	51
	x10d inadequate police/patrol supervision	42
	x10e issuance of driver's licenses/ operating permits to unqualified drivers	2
	x10f inadequate vehicle management	13
	x10g inadequate site monitoring	45
X11 resource management	x11a lack of funding	0
	x11b human resource staffing/manning	1
	x11c road infrastructure/vehicle design discrepancy	4
X12 organizational climate	x12a ineffective rules and regulations	14
	x12b lack of awareness of traffic safety	32

5. Critical failure routes analysis

5.1. Bayesian Network analysis

In this study, "failure" refers to a systemic breakdown in safety–critical processes that results from the interaction of latent organizational, supervisory, and operational deficiencies. "Failure Routes" are causal pathways that describe how latent systemic weaknesses propagate through a system to result in operational failures (Reason, 1990). To identify failure routes of sub-categories across four levels, we selected significant causal relationships with high odds ratio (OR) and the Bayesian Network primarily focuses on variables with high statistical significance.

To analyze the relationship between the upper and lower level sub-categories OR and 95 % confidence interval (*CI*) were calculated as **Equation 4–7**. OR provides a special and clear explanation of the ideal value to analyze multidimensional factors.

$$OR = \frac{a/c}{b/d} = \frac{a*d}{b*c} \tag{4}$$

$$CI_{95\%} = (Lowerlimit, Upperlimit) \tag{5}$$

$$Lowerlimit = e^{(\ln(OR)-1.96*\sqrt{\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}})} \tag{6}$$

$$Upperlimit = e^{(\ln(OR)+1.96*\sqrt{\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}})} \tag{7}$$

where a is number of cases where the lower level sub-category and the higher level sub-category are present. b is number of cases where the lower level sub-category is absent and the higher level sub-category is present. c is number of cases where the lower level sub-category is present and the higher level sub-category is absent. d is number of cases where the lower level sub-category and the higher level sub-category are absent.

This study directly adopts the HFACS-CMV four levels network structure. Variables showing significant causal relationships between the sub-categories of the upper and lower level sub-categories were incorporated into the BN computation. The BN modeling was done using GeNIe software. BN relies on Bayes' theorem to propagate information between nodes, **Equation (8):**

$$P(Y|X) = \frac{P(X|Y)*P(Y)}{P(X)} \tag{8}$$

where $P(Y)$ is prior probabilities, $P(X|Y)$ is the conditional probability (the likelihood of the evidence given the hypothesis), and $P(Y|X)$ is the posterior probability of the hypothesis (the likelihood of Y being in a certain state, conditional on the evidence provided).

In this study, the coding results of 100 crash reports were used as prior probabilities. When the probabilities for certain low-level sub-categories were set to 100 %, Bayesian theory was applied to update these probabilities, resulting in posterior probabilities. The sub-categories at higher levels with the highest posterior probabilities or the most significant changes were identified to construct the critical failure routes.

5.2. Relationships between contributing factors

We calculated the OR and 95 % *CI* for the sub-categories across different levels to clarify their relationships. An OR greater than 1 indicated that the sub-category at the upper level influences the sub-category lower level. If the *CI* did not include 1, the OR was considered statistically significant. Given the large number of sub-categories, we reported only the ORs and lower limits of the 95 % *CIs* that were greater than 1 (as Appendix C).

The relationship between **Level 4** Organizational Influences and **Level 3** Unsafe Supervision illustrated that significant associations between poor company supervision and government oversight in various sub-categories. The ineffective implementation of rules and regulations, and inadequate vehicle management, significantly impacted the inadequate pre-departure safety inspection, with OR values as high as 36.84 and 24.21, respectively. Additionally, a poor organizational climate heightened the likelihood of insufficient safety training for employees, failure to rectify and reinspect safety hazards, and the occurrence of improper staffing. However, resource management had not demonstrated a significant relationship with unsafe supervision, which might be due to the limited elaboration on resource management details in crash reports.

The relationship between **Level 3** Unsafe Supervision and **Level 2** Preconditions for Unsafe Acts illustrated that the failure to identify an at-risk vehicle significantly impacted the vehicle issues, with the highest OR reaching 26.23. Fatigue driving was significantly influenced by the failure to rectify and reinspect safety hazards, as well as by conducting transportation tasks not in accordance with rules and regulations, with the OR exceeding 10. Additionally, Failure to address known issues could exacerbate adverse driver conditions and poor technical environments involving vehicles and roads, particularly in cases of drug driving, fatigue driving, and vehicle malfunctions. Results indicated no significant causal relationship between the physical environment and unsafe supervision.

The relationship between **Level 2** Preconditions for Unsafe Acts and **Level 1** Unsafe Acts**,** and the relationship between **Level 3** Unsafe Supervision and **Level 1** Unsafe Acts are shown in Appendix B. The weather-related road conditions significantly impacted the failure to yield, with OR reaching 5.44. Illegal operation significantly was influenced by transportation tasks not in accordance with rules & regulations and failure in establish/implement safety production systems with OR exceeding 9. Failure to identify at-risk drivers, companies' inadequate supervision, and vehicle issues could lead to errors. Failure to address known issues and substandard operator conditions could exacerbate the prevalence of exceptional violations. The inadequate supervision of companies and planned inappropriate operations were related to routine violations.

5.3. Critical failure routes of key error

This study focused on the sub-categories with the highest frequency error: wrong responses to emergencies. Fig. 4a illustrates the paths with prior probabilities of x1b (wrong response to emergency). Given the condition of sub-category x1b with a probability of 100 %, the probability of x5a (weather-related road conditions) increased by 8 % (Fig. 4b). Furthermore, when x1b and x9a (inadequate pre-departure safety inspection) were both at a probability of 100 %, x10a (ineffective implementation of rules and regulations) and x10c (inadequate supervision of companies) shown high probabilities of 46 % and 47 %, respectively (Fig. 4c). Thus, the critical failure routes to x1b were identified as x10a → x9a → x1b, x10c → x9a → x1b, and x5a → x1b.

5.4. Critical failure routes of key violations

This study focused on the sub-categories with the highest frequency violations: exceptional violations (over speeding) and routine violations (invalid licenses). Fig. 5a depicts the paths with prior probabilities for x3c (invalid license). Given the condition of x3c with a probability of 100 %, the probability of x7c (insufficient safety training for employees) increased by 21 % (Fig. 5b). Moreover, when both x3c and x7c were at a probability of 100 %, x10c (inadequate supervision of companies) and x10g (inadequate site monitoring) shown high probabilities of 54 % and 51 %, respectively. The probabilities of x10c, x10g, and x12b (lack of awareness of traffic safety) increased by 6 %, 8 %, and 5 %, respectively (Fig. 5c). Thus, the critical failure routes to x3c were identified as x10c → x7c → x3c and x10g → x7c → x3c.

The paths for x2a (over speeding) with prior probabilities are shown in Fig. 5a. Given the condition of x2a with a probability of 100 %, the probability of x4e (fatigue driving) increased by 1 % (Fig. 6a**)**. Additionally, when both x2a and x4e were at a probability of 100 %, x7f (inadequate on-route monitoring) shown the highest probability of 48 %, while the probability of x9c (transportation tasks not in accordance with rules and regulations) increased the most by 24 % (Fig. 6b). Fig. 6c demonstrates that when x2a, x4e, and x7f were at a probability of 100 %, x10c (inadequate supervision of companies) shown the highest probability of 71 % and increased by 14 %. Fig. 6d shown that when x2a, x4e, and x9c were at a probability of 100 %, x10c also shown the highest probability of 60 %, while x12a (ineffective rules and regulations) increased the most by 17 %. Thus, the critical failure routes to x2a were identified as x10c → x7f → x4e → x2a, x10c → x9c → x4e → x2a, and x12a → x9c → x4e → x2a.

6. Discussion

This study established a HFACS-CMV, analyzing the primary causal factors and critical failure routes based on 100 investigation reports of 10 or more fatalities crashes in China. By conducting quantitative in-depth analyses of these causes of crashes and applying system safety

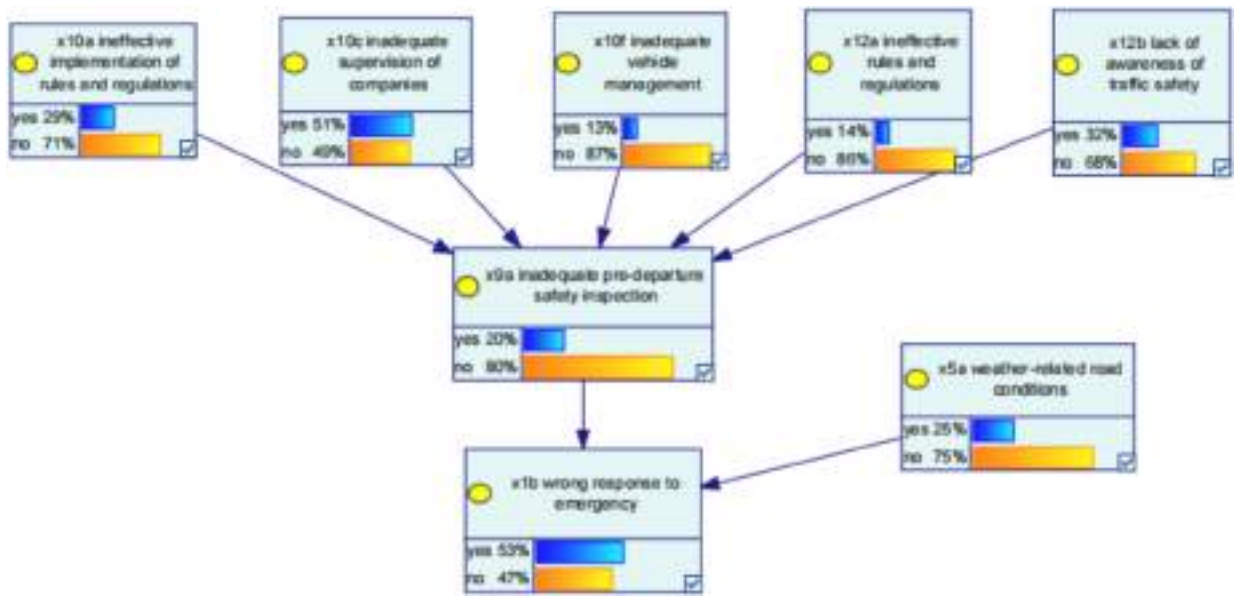

Fig. 4a. Prior Probabilities of x1b Wrong Response to Emergency.

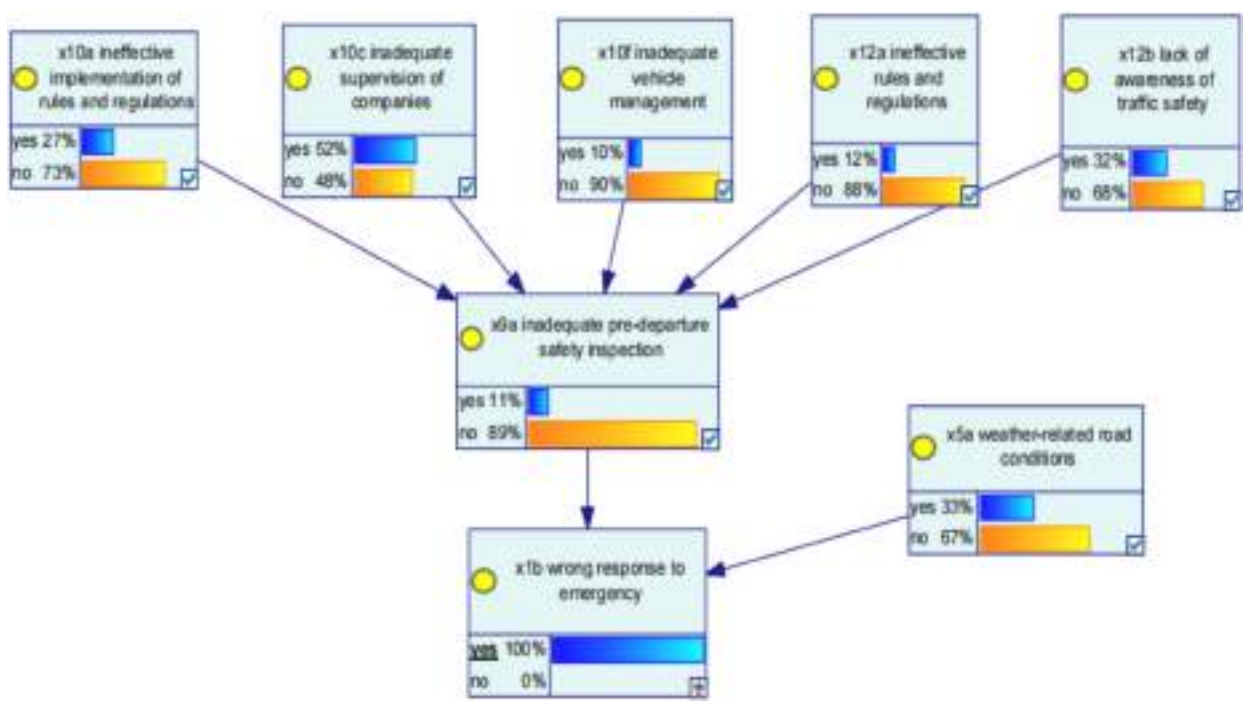

Fig. 4b. Posterior Probability under 100% x1b Conditions.

management theory, we aim to shift road traffic safety governance towards proactive prevention.

6.1. *Implications for In-depth investigation of road traffic crashes*

The recommendations derived from post hoc analysis of crash data are limited by the current road crash reporting protocols. The crash reports used were not designed with HFACS in mind, consequently detail, particularly of higher-level causses can be lacking. The lack of information on unsafe supervision and organizational influences in some reports underscores the need for more consistent and comprehensive reporting on higher-level factors. Standardizing crash investigation procedures is crucial. The DaCoTA project's investigation manual can assist countries in conducting thorough crash investigations (Talbot et al., 2024). Additionally, continuous review and updating of crash report forms are necessary to ensure they reflect the full range of contributing factors (Rolison et al., 2018). Gooding and Box argue for dedicated, safety-oriented road crash investigations with independent status, similar to those for rail, air, and maritime crashes (Gooding & Box, 2022). Establishing specialist units focused on identifying causes and recommending preventative measures, rather than apportioning blame, is essential for advancing road safety (Gooding & Director, n.d.). Nonetheless, it is possible to develop and evidence effective use of the new HFACS-CMV approach. Future research should test this approach with crash reporting mechanisms which are informed by a safe system approach.

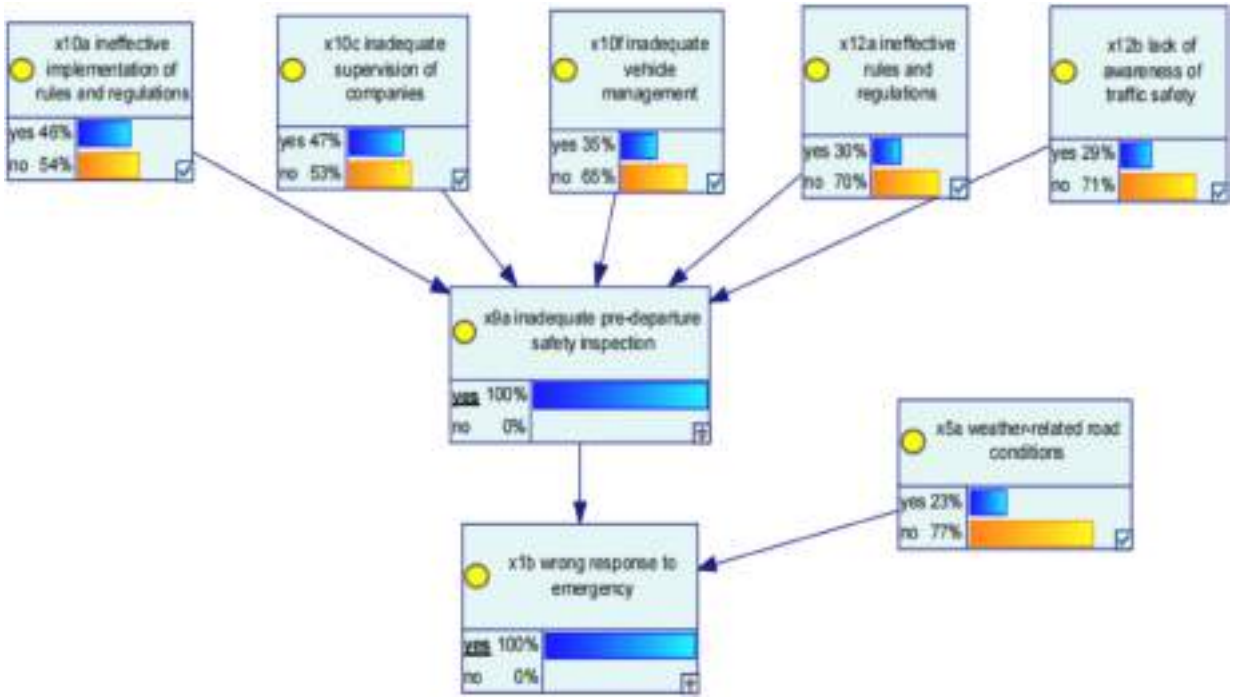

Fig. 4c. Posterior Probability under 100% x1b and x9a Conditions.

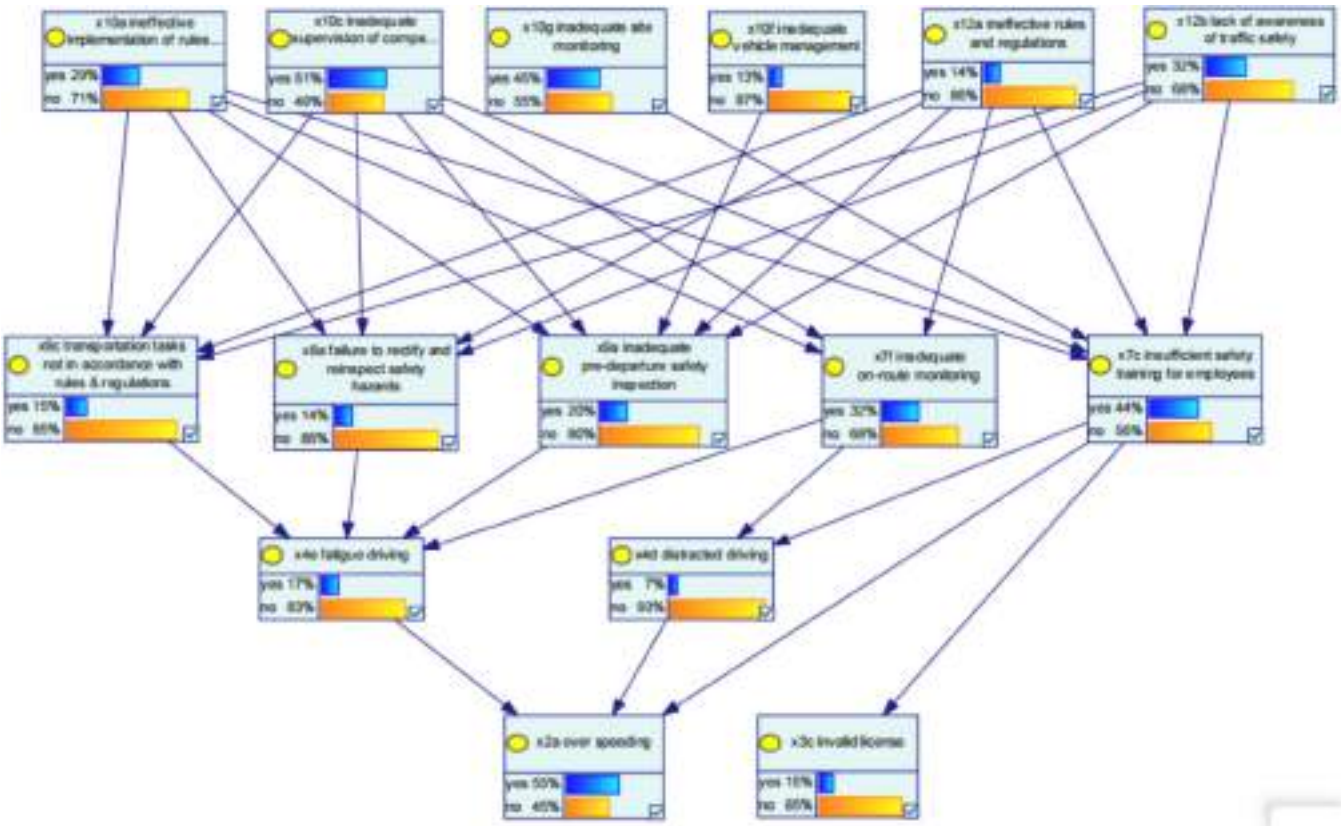

Fig. 5a. Prior Probabilities of x3c Invalid License and x2a Over Speeding.

6.2. Key error and Improvement strategies

For factors with high frequency or severity, corresponding strategies should be adopted to enhance road safety. The results confirm that weather-related road conditions significantly increase the likelihood of incorrect emergency responses, which is consistent with existing research (Wei et al., 2024). This highlights the urgent need for enhanced training in defensive driving skills for drivers in China, particularly focusing on emergency preparedness. Effective educational and management measures to improve drivers' emergency response skills are

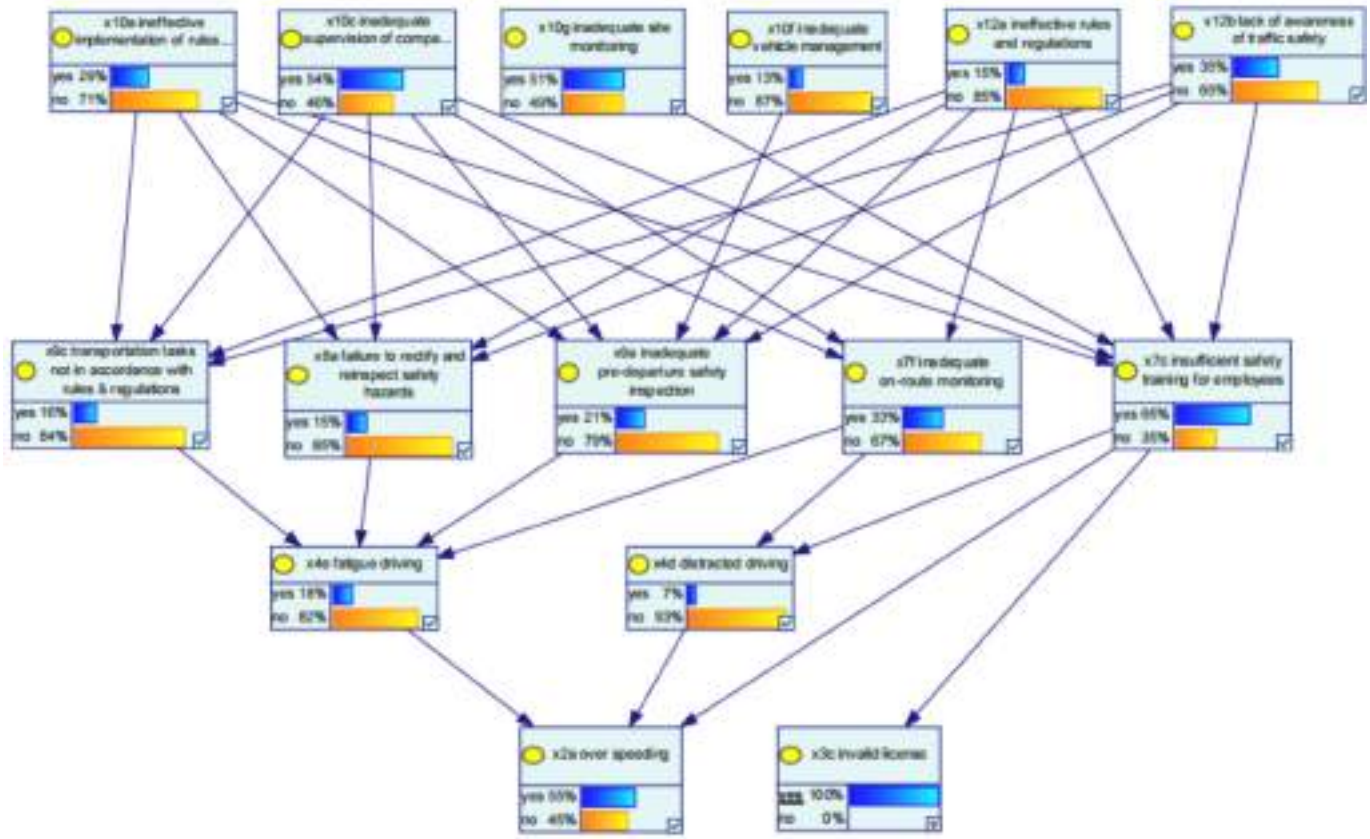

Fig. 5b. Posterior Probability under 100% x3c Conditions.

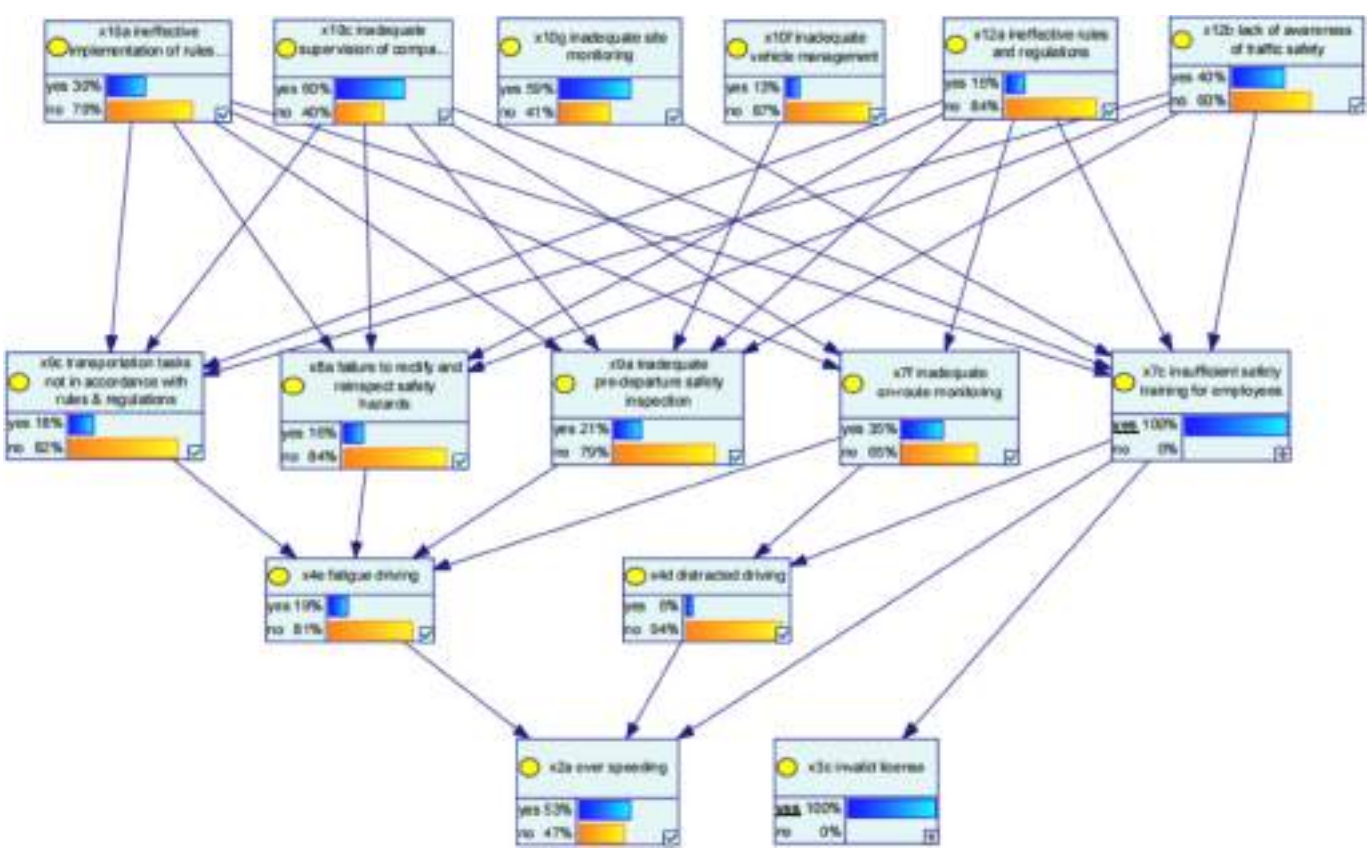

Fig. 5c. Posterior Probability under 100% x3c and x7c Conditions.

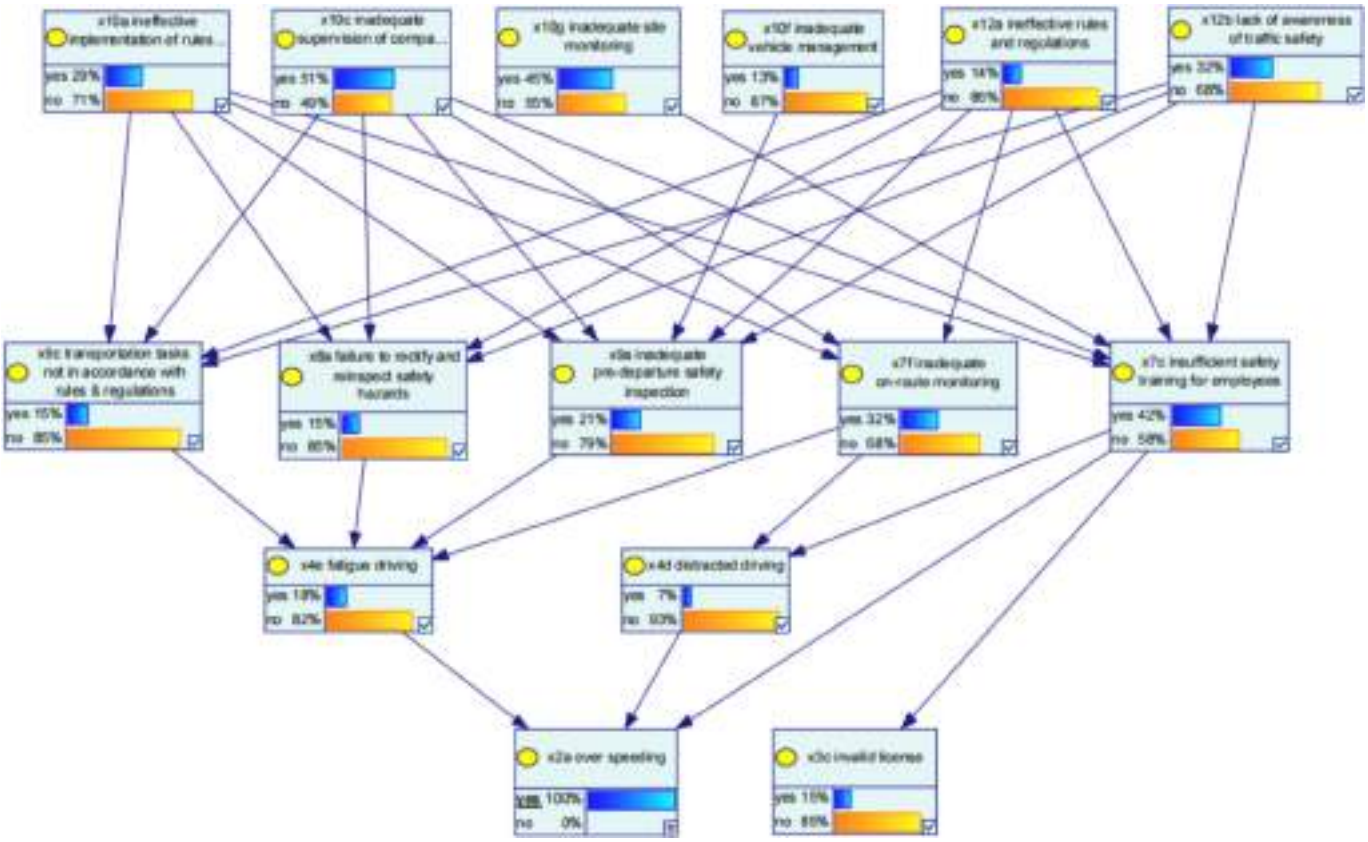

Fig. 6a. Posterior Probability under 100% x2a Conditions.

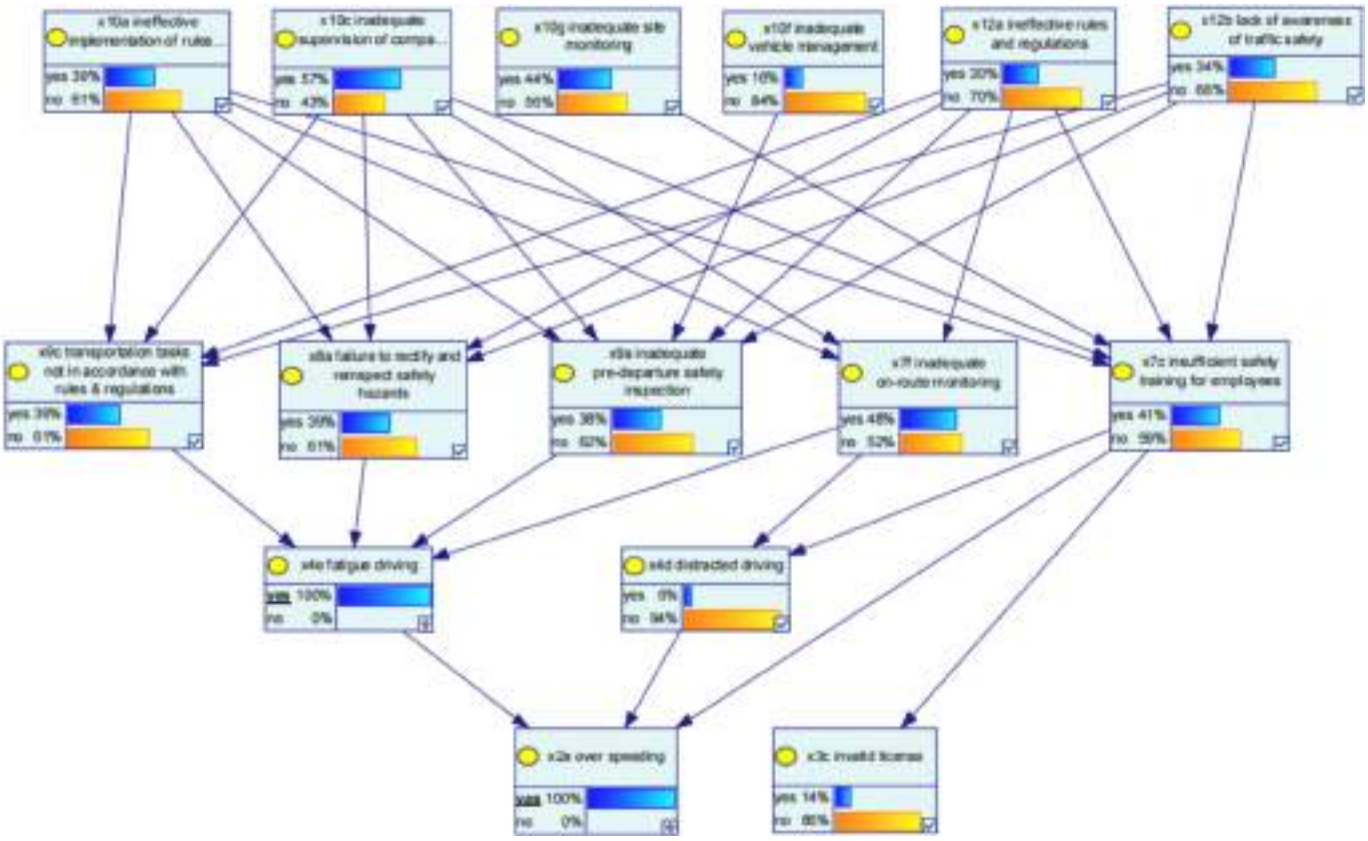

Fig. 6b. Posterior probability under 100% x2a and x4e conditions.

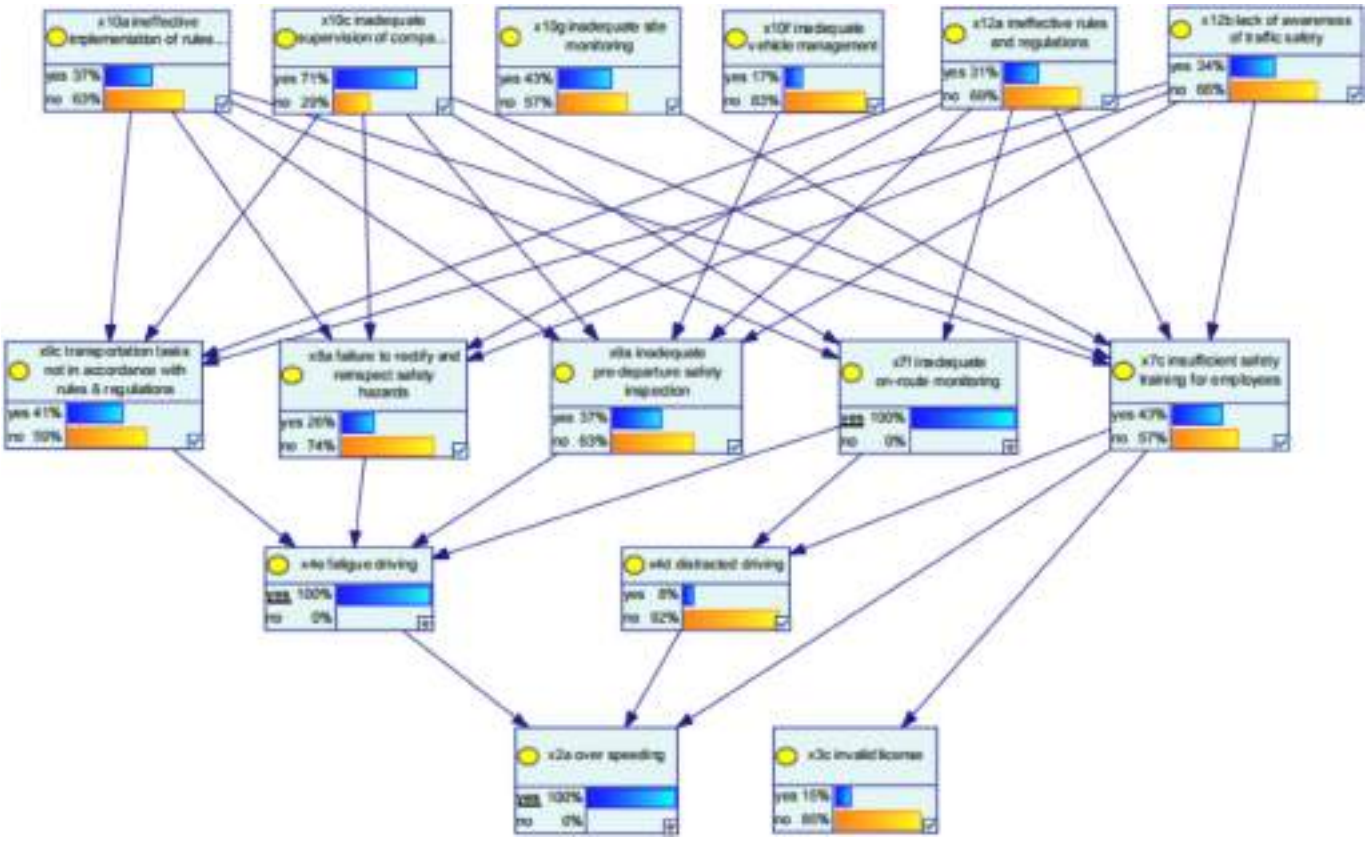

Fig. 6c. Posterior probability under 100% x2a, x4e and x7f conditions.

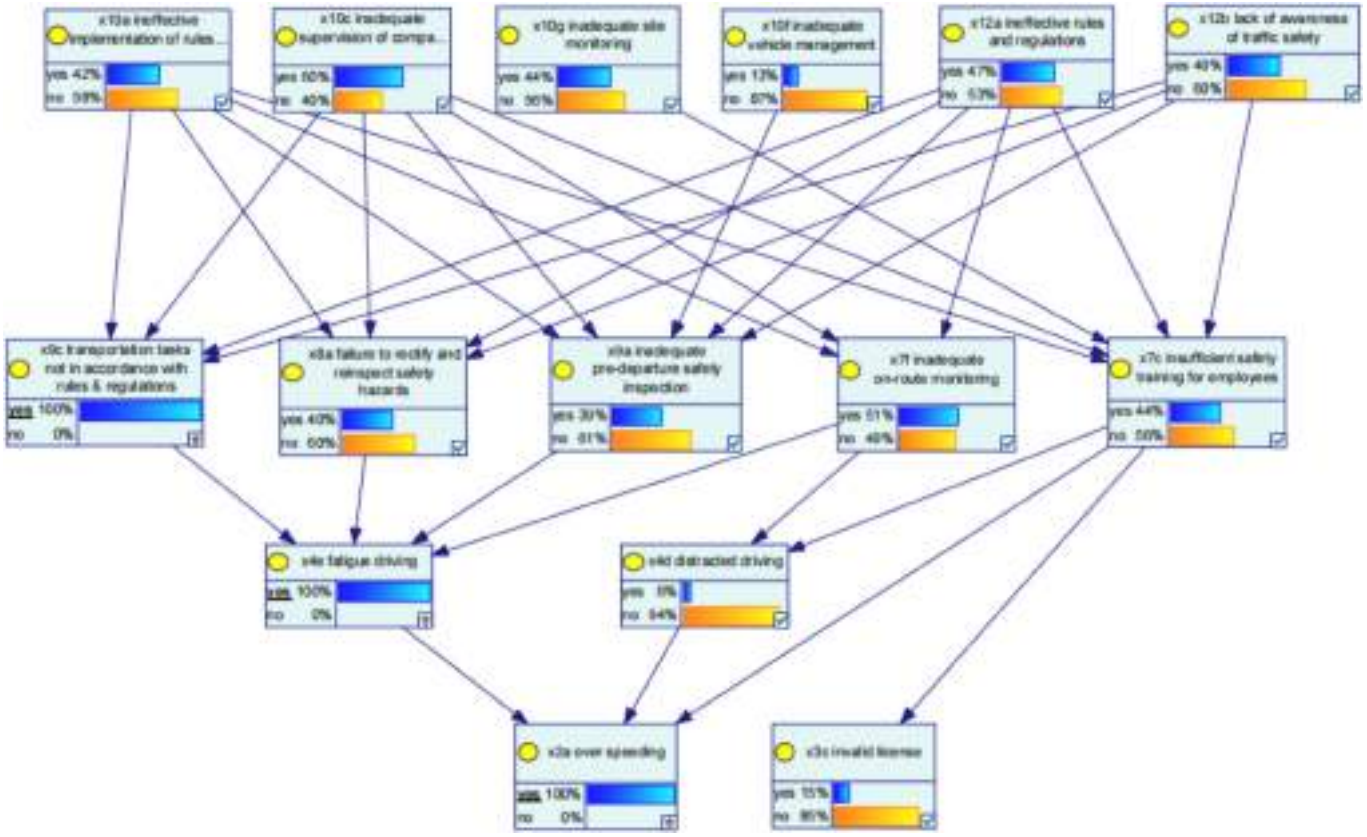

Fig. 6d. Posterior probability under 100% x2a, x4e and x9c conditions.

crucial for reducing incorrect behaviors (Horswill et al., 2021). Moreover, in emergency situations, individuals often struggle to respond correctly due to deficiencies in perceptual reactions and specific neural patterns of sensory perception (Haufe et al., 2011). In-vehicle Advanced Driving Assistance Systems, which integrate vehicle active safety and driver assistance technology, can mitigate driving risks in emergencies. For instance, the Automatic Emergency Braking system has been proven to significantly reduce crashes (Yurtsever et al., 2020), and Forward Collision Warning systems alert drivers to impending collisions (Shao et al., 2023). Simultaneously, investing in infrastructure to ensure its resilience against weather conditions is crucial. For example, increasing road surface friction, installing sensors and systems to monitor weather conditions aid in real-time road management, enabling timely salting or snow removal.

Our study also found that ineffective implementation of rules and regulations, along with poor company supervision, leads to insufficient pre-departure safety inspections, thereby increasing the likelihood of incorrect emergency responses. Although China has established operational procedures for road transport drivers, several shortcomings remain. Specifically, the lack of specific and quantifiable regulations regarding drivers' physiological and psychological conditions is a primary issue. Implementing psychological scales or wearable devices, such as wristbands, can enhance the accuracy of detecting and evaluating drivers' mental and physical states (He et al., 2024). Furthermore, the rigor of pre-departure inspections varies by region, company, and individual, presenting a challenge for regulatory agencies to effectively supervise each vehicle. Leveraging information technology for coordinated management could improve this process.

6.3. Key violations and improvement strategies

Numerous studies have consistently shown that violations significantly contribute to crash involvement (Z. Zhang et al., 2019). The results suggest that insufficient safety training correlates with a higher incidence of invalid licenses. Regular, comprehensive safety training and continuous education are essential to make drivers cognizant of the risks associated with violations, thereby reducing their frequency (Z. Zhang et al., 2019). Furthermore, companies should maintain detailed records of all driver training activities, with a particular focus on drivers with previous illegal driving behaviors to monitor and improve their compliance. Poor company supervision, ineffective regulations, and a lack of traffic safety awareness are fundamental issues leading to unsafe acts. Enhancing company supervision and addressing organizational deficiencies, particularly in safety systems, training, and inspections, are critical steps. For instance, common causes of inadequate supervision include the failure to establish or implement safety production systems, insufficient safety training, and inadequate vehicle inspections. Implementing a comprehensive Safety Management System (SMS), which includes hazard identification, occurrence reporting, risk management, performance measurement, and quality assurance, can significantly reduce crashes.

Secondly, the results indicate that inadequate drivers' status on-route monitoring elevates the probability of fatigue driving, which in turn substantially augments the risk of speeding violations. Research supports that real-time monitoring technologies that assess drivers' real-time capability to perform tasks by collecting observable data (Hayley et al., 2021). These systems can monitor drivers' mental states and behaviors, issuing alerts when distractions or fatigue are detected (Michelaraki et al., 2023). Hence, it is imperative for transportation enterprises to adopt real-time monitoring technology to effectively supervise driver behavior and provide timely warnings to prevent unsafe practices. Moreover, previous studies have shown that crashes are also related to the alignment and condition of the road network (Abegaz et al., 2014). For instance, Kong et al. (2020) found that driving on roads with higher functional classifications is strongly associated with longer speeding durations. Rossi et al. (2011) highlighted the negative impact of monotonous driving environments on driving performance. Therefore, improving early warning systems for hazardous road sections, enhancing enforcement, and upgrading conditions at crash-prone locations are crucial for enhancing road safety.

Thirdly, poor site monitoring and insufficient police/patrol supervision contribute significantly to road traffic crashes. These issues highlight the persistent lack of control and enforcement regarding traffic regulations, such as speeding, driving under the influence, and failing to respect other road users' rights (Goniewicz et al., 2016). Further effort needed as well to monitor speed control strategies like; using the radar control and physical speed restraint measures (i.e., rumble strips) (Abegaz et al., 2014). Research shows that strict enforcement measures can significantly improve driver behavior and reduce crashes and fatalities (Lv et al., 2022). Smart enforcement combines pre-risk assessments to target enforcement actions, and post-enforcement strategies are adapted to the behaviors. The implementation of law could be improved by using smart enforcement techniques (Blanc & Faure, 2020).

Additionally, transportation tasks not conducted in accordance with rules and regulations lead to an increased frequency of fatigue driving. For instance, failure to comply with the requirement of taking a 20-minute break after continuous driving for 4 h. The NTSB has identified several factors—including scheduling policies, hours of service, time off between assignments, and circadian disruptions—as critical contributors to driver fatigue (Marcus & Rosekind, 2017). By considering working hours, workload, and circadian rhythms, biomathematical models can optimize driving schedules and tasks, significantly enhancing driving safety and reducing fatigue-related crashes. Biomathematical models are extensively utilized in civil aviation to forecast alertness and optimize scheduling, which could be adapted for road traffic (Knock et al., 2021).

6.4. Limitations and future research direction

First, differences in reporting practices across regions and years may introduce variability in the data and the focus on crashes with ten or more fatalities in China, while providing valuable insights into high-risk scenarios, may limit the broader applicability of the findings to other contexts or less severe crashes. Secondly, the reliability of the HFACS-CMV framework was assessed by a panel of five transportation experts; while this provides initial validation, expanding the number and diversity of experts in future research could strengthen the generalizability of the findings. The observed variability in frequencies across HFACS sub-categories suggests potential data imbalances, which could be further investigated in future studies to enhance methodological robustness. Thirdly, this study emphasizes human and management factors, lacking a quantitative analysis of the impact of road geometric factors on driver behavior and crashes. Since different road types, crash patterns, and the specific characteristics and causes of crashes involving passenger and freight vehicles may vary, future studies should aim to collect more data for a more refined causal analysis of crashes. While the HFACS-CMV framework is effective in identifying causal factors, integrating real-time data and advanced analytics could enhance predictive capabilities and intervention strategies. Future research should explore the use of emerging technologies, such as machine learning and big data analytics, to better predict and prevent crashes.

7. Conclusions

The HFACS-CMV framework provides a systematic and reliable tool for identifying the human factors contributing to commercial vehicle crashes, addressing critical gaps in traditional investigative methods. By integrating both behavioral and organizational perspectives, this framework uncovers systemic issues, such as inadequate supervision, ineffective regulations, and poor organizational practices, which are often overlooked. These findings highlight the need for enhanced

reporting protocols and standardized investigation procedures to capture deeper, underlying causal factors.

This study applied HFACS-CMV to quantitatively analyze 100 crashes, identifying common causal patterns. The results emphasize the critical role of unsafe driver behaviors, such as improper emergency responses, speeding, and driving without valid licenses. Furthermore, the HFACS-CMV with BN model reveals a crucial failure route: inadequate government oversight that leads to poor company supervision and operational planning, ultimately resulting in unsafe operator conditions and behaviors. These insights provide a foundation for targeted interventions to break the chain of failures. To enhance road safety, this study recommends that transportation companies and government agencies implement regular safety training, stricter regulatory enforcement, and more robust operational planning.

In summary, this study critically examines the systemic and behavioral factors contributing to road traffic crashes, providing actionable insights into critical failure routes and targeted mitigation strategies. A holistic, multifaceted approach—combining behavioral interventions, organizational improvements, and technological innovations—is essential to reducing crashes and fatalities. By fostering a culture of safety and advancing analytical methodologies, this research contributes to a safer and more sustainable transportation sector.

8. Author statement

The authors confirm contribution to the paper as follows: study conception and design: Xuesong Wang and Yanru Zhou; data collection: Yanru Zhou, Xiaowei Tang and Shikun Liu; analysis and interpretation of results: Yanru Zhou; draft manuscript preparation: Xuesong Wang, Yanru Zhou, Ashleigh Filtness, Chao Wang and Zhicheng Wang. All authors reviewed the results and approved the final version of the manuscript.

CRediT authorship contribution statement

Xuesong Wang: Writing – review & editing, Funding acquisition. **Yanru Zhou:** Writing – review & editing, Writing – original draft, Resources, Methodology, Formal analysis, Data curation. **Ashleigh Filtness:** Writing – review & editing. **Chao Wang:** Writing – review & editing. **Xiaowei Tang:** Writing – original draft, Data curation. **Shikun Liu:** Data curation. **Zhicheng Wang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.aap.2025.108047.

Data availability

Data will be made available on request.

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